

AI technologies

1/ Designing AI solutions

Presentation 1:

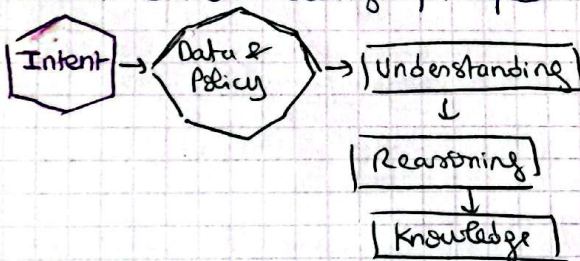
- Effective design is the key to unlocking the full potential of AI
- Well-designed AI sols have a significant impact on user exp, workflow efficiency, business growth
- User-centric design ensures that AI sols are intuitive and easy to use, enhancing user exp
- Thoughtfully designed AI sols streamline complex processes, automate tasks, and improve productivity
- Design-led organizations have a competitive edge
- Exceptional design fosters customer loyalty & satisfaction

Key principles of designing AI sols

usability accessibility scalability ethics

AI design thinking:

structured approach to designing AI products and experiences that prioritize human-centered design principles



Workflow:

Systematic series of steps or processes involved in the creation, development, deployment & maintenance of AI system

- Define the product vision
- Set the right expectations
- Plan a product strategy
 - ↳ Agile Lean startup methodology



- Discovery stage
MVP: Min Viable Product
- Validation stage
MVT: MV Tests

- Scaling stage and let users give feedback
- Create a tech and infrastructure strategy

2/ AI frameworks

Comprehensive software libs / tools designed to facilitate the development, implementation and deployment of AI models / apps

ML frameworks
neural network-centric

DL frameworks
ML algos

Types of AI frameworks:

- General-purpose AI
- DL
- ML
- NLP
- Computer vision
- Reinforcement Learning
- AutoML
- Distributed AI
- Quantum machine Learning

Popular AI frameworks

- TensorFlow (DL, ML, TFX)
- PyTorch (NN, research, NLP, reinforcement Learning)
- Keras (NN)
- Scikit-learn (ML)

Challenges:

- Interoperability
- Resource intensiveness
- Explainability
- Data privacy
- Bias, Fairness
- Real-time processing

3/ Ethics in AI

- Privacy vs surveillance
- Human employment vs AI automation
- Autonomy vs control in autonomous vehicle decision-making
- Security vs accessibility in AI data usage

How to avoid ethical issues:

- Establish ethical guidelines and international standards
- Promote transparency / explainability
- AI impact assessments

Deepfakes

Recreation of (fake) images, voice using AI

Facenet:

also for face recognition

→ similarity score

score < threshold → deepfake

Bias in AI:

Presence of systematic/unfair preferences in data/algorithms/AI systems
→ discriminatory outcomes

- Implicit bias
- Sampling "
- Temporal "
- Overfitting to training data
- Edge cases & outliers
- User interaction bias

COMPAS algo:

Correctional Offender Management
Profiling for Alternative Sanctions

ML → predict the potential for
criminal defendants to reoffend

bias for people of color

Amazon's recruiting algo
recruiting men over women

Bias metrics

Equalized Odds Disparate Impact Non-Parametric cohort analysis

LLM prompt engineering

NLP: Natural Language Processing
subset of AI that enables machines to comprehend, interpret and generate human-like text

LLM: Large Language Models
trained on massive amounts of data (text)

Zero-shot prompting
describe a task/
desired output

One-shot
provide exple

few-shot
small nbr of exples

chain of thought
complex tasks → series of intermediate steps
output for each step

5% Future directions & recent trends

Advancements in ML:

- Unsupervised ML
- Transfer Learning
- Explainable AI
- DL (NN, Img/speech recognition, NLP)
- Reinforcement Learning

Emerging technologies:

- LLM
- LVM (Large vision Model)
- Auto ML / Auto AI
- ML lib on Spark